A MULTI-VIEW RETRIEVAL SYSTEM

MULTI-FACETED MUSIC RETRIEVAL.

Combining audio, text, and joint embedding into a single recommendation system that outperforms any single view.

Team

Wenny · Sid · Issac · MJ · Jiayi · Yunchu

Dataset

Free Music Archive · 2,000 tracks

/ 02 — ARCHITECTURE

From raw audio & metadata to fused recommendations.

Overall Architecture & Web Demo — Issac

CLAP Embeddings & Results — Jiayi

Acoustic Similarity with OpenL3 — Wenny

Semantic Search with SBERT & Lyrics — Sid

Evaluation Framework & Fusion Analysis — Yunchu (Helena)

Bonus Experiment : Audio-to-Spectrograph Analysis — MJ

This project builds a **multi-view retrieval system** over the Free Music Archive (FMA) dataset, an open-licensed benchmark of 8,000 tracks across 8 top-level genres.

Each view produces an independent embedding space, and a fusion layer combines them to give a combine score for recommendation.

QUERY · MOOD

"Chill lo-fi vibes"

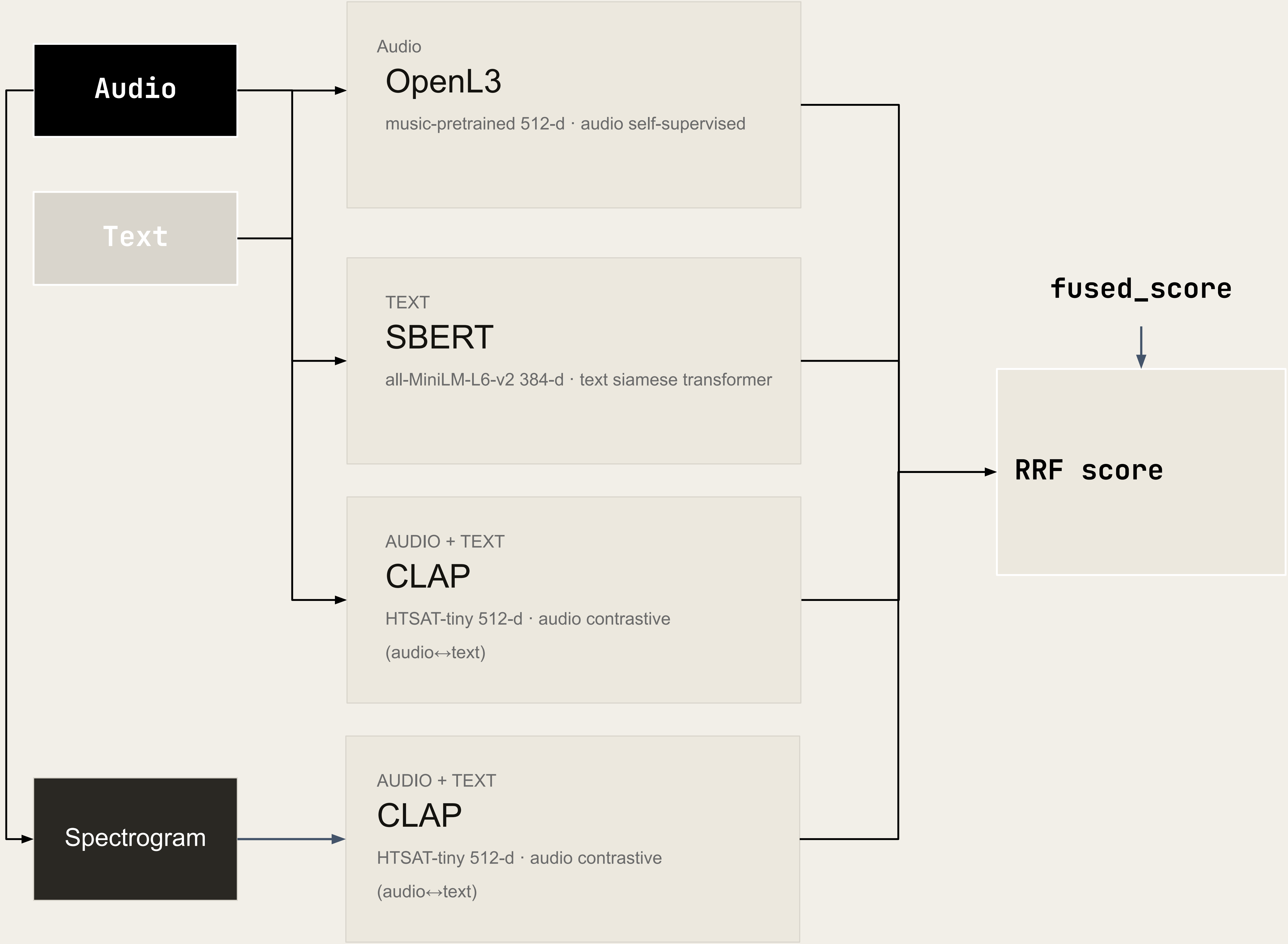
requires acoustic features — mood & instrument feel

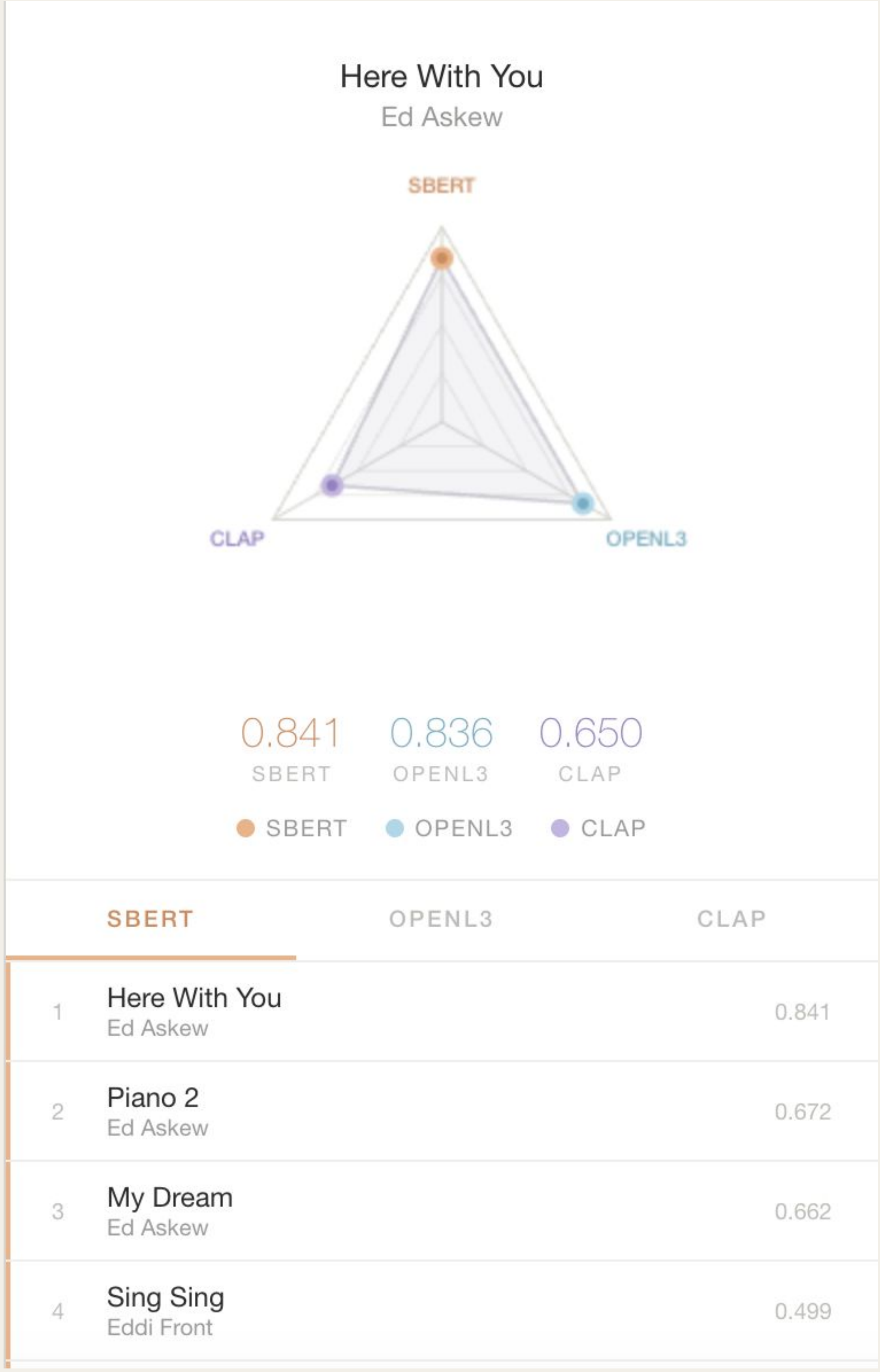
QUERY · SEMANTIC

"Folk artists about travel"

requires text & lyrics — semantic meaning

"We use three different embedding models to search for similar music — each one captures a different dimension of similarity: sound, meaning, and vibe.”





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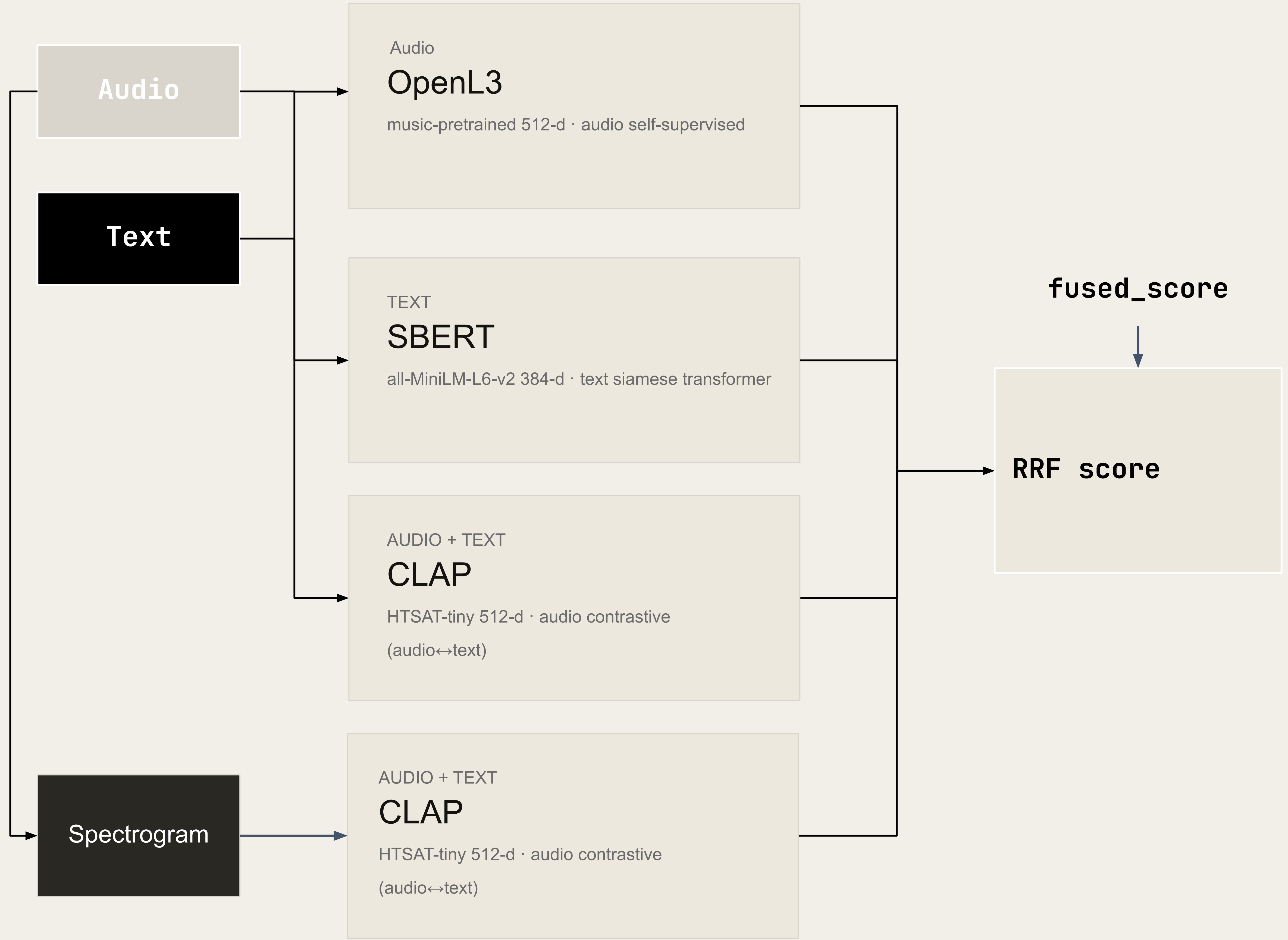
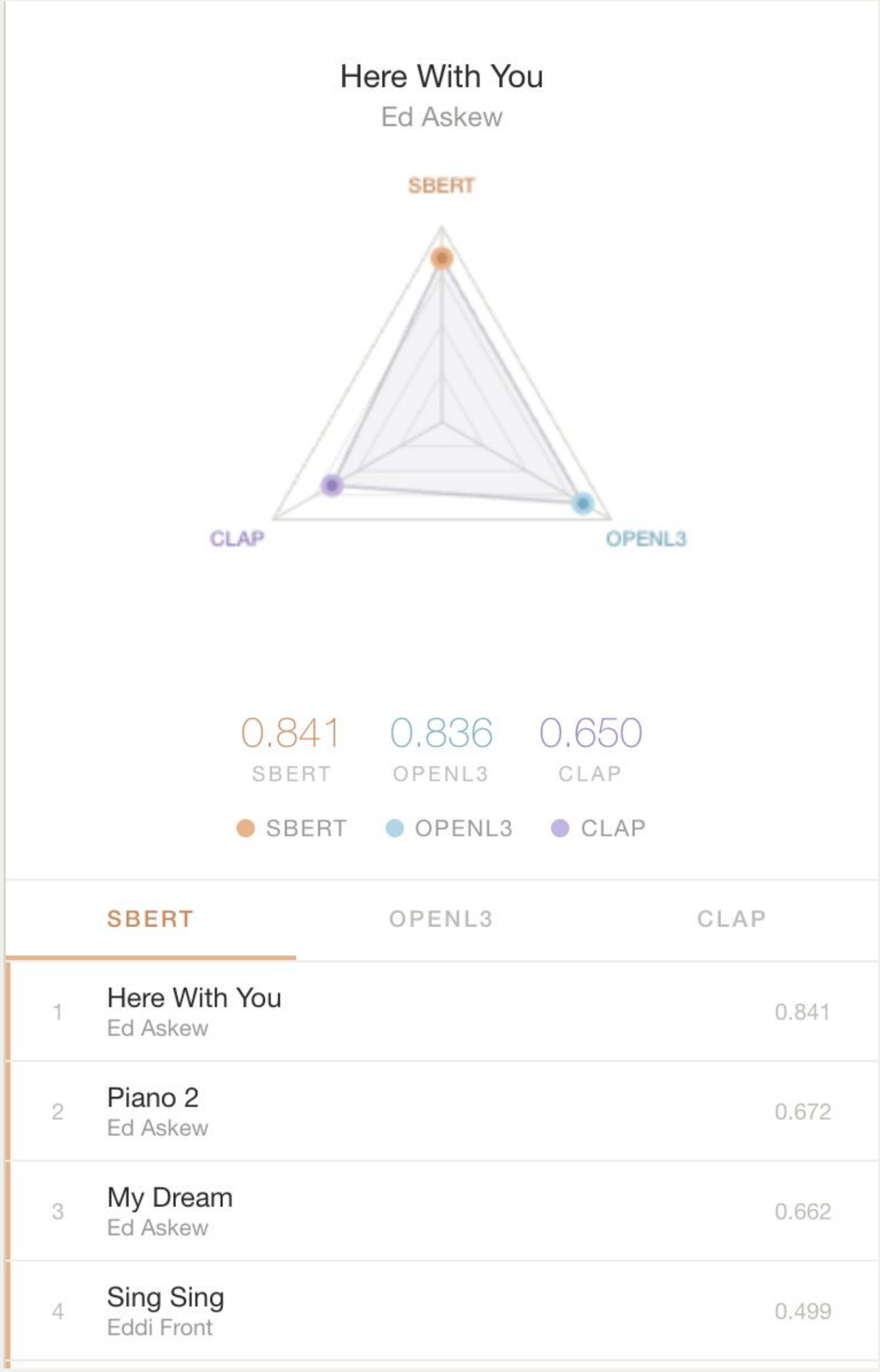
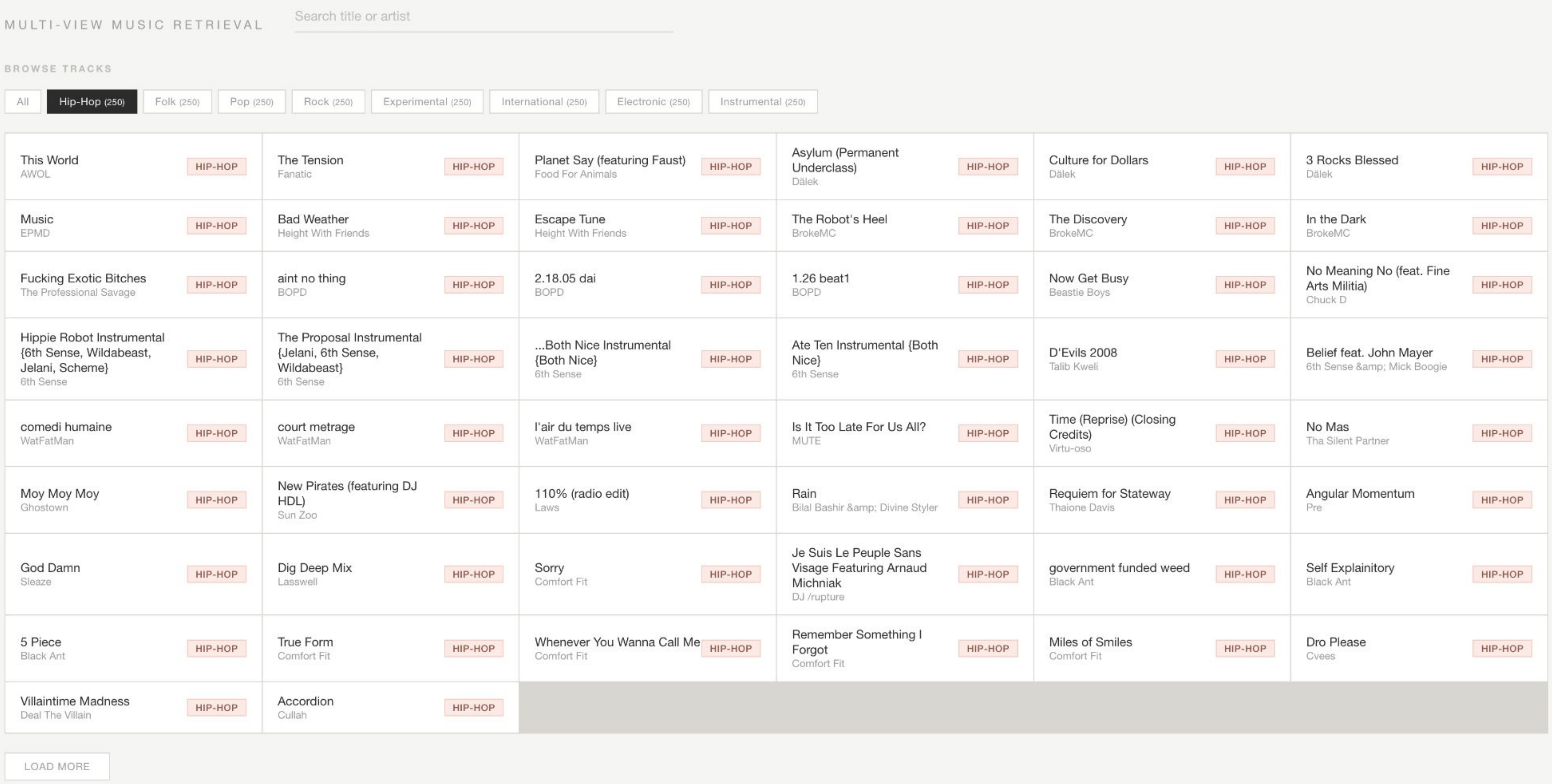




/ SECTION 01

Demo.

Here’s what the final outcome will look like, before we do a deep dive



Genre is derived from FMA metadata.

/ SECTION 01

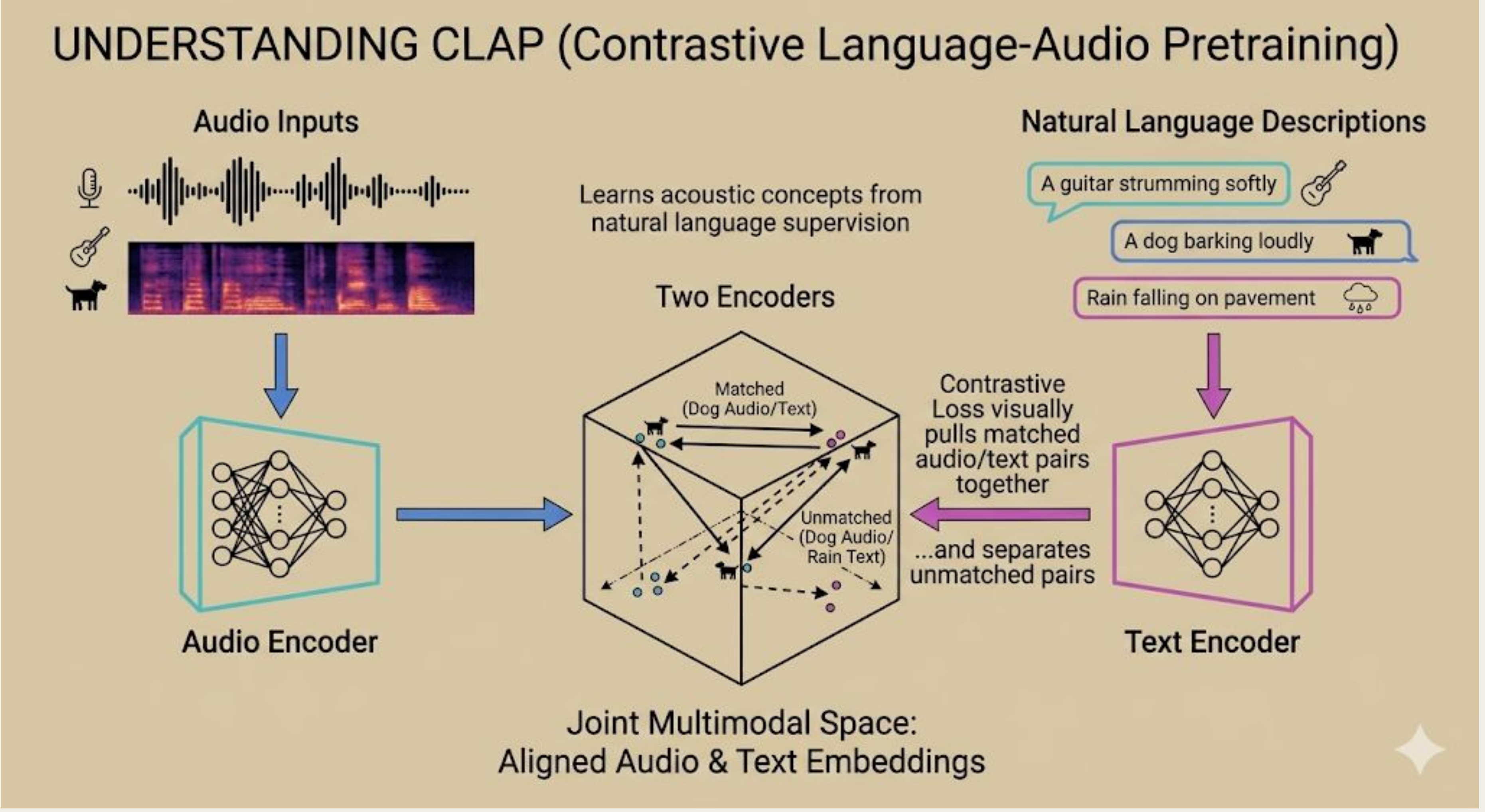
CLAP.

Vibe & text-to-music search — a 512-d shared space for audio and text via contrastive pretraining.

/ 01 — CLAP RESULTS

CLAP - Contrastive Language Audio-Pretraining

- Learns acoustic concepts from natural language supervision
- Two encoders
- Contrastive loss
- Audio and text descriptions -> joint multimodal space



Pipeline:

```
Audio file (MP3) → Resample to 48 kHz → Clip to 10s → HTSAT-tiny → 512-d vector → L2-normalize
```

CLAP retrieves on-genre top results across all natural-language queries — even with no fine-tuning.

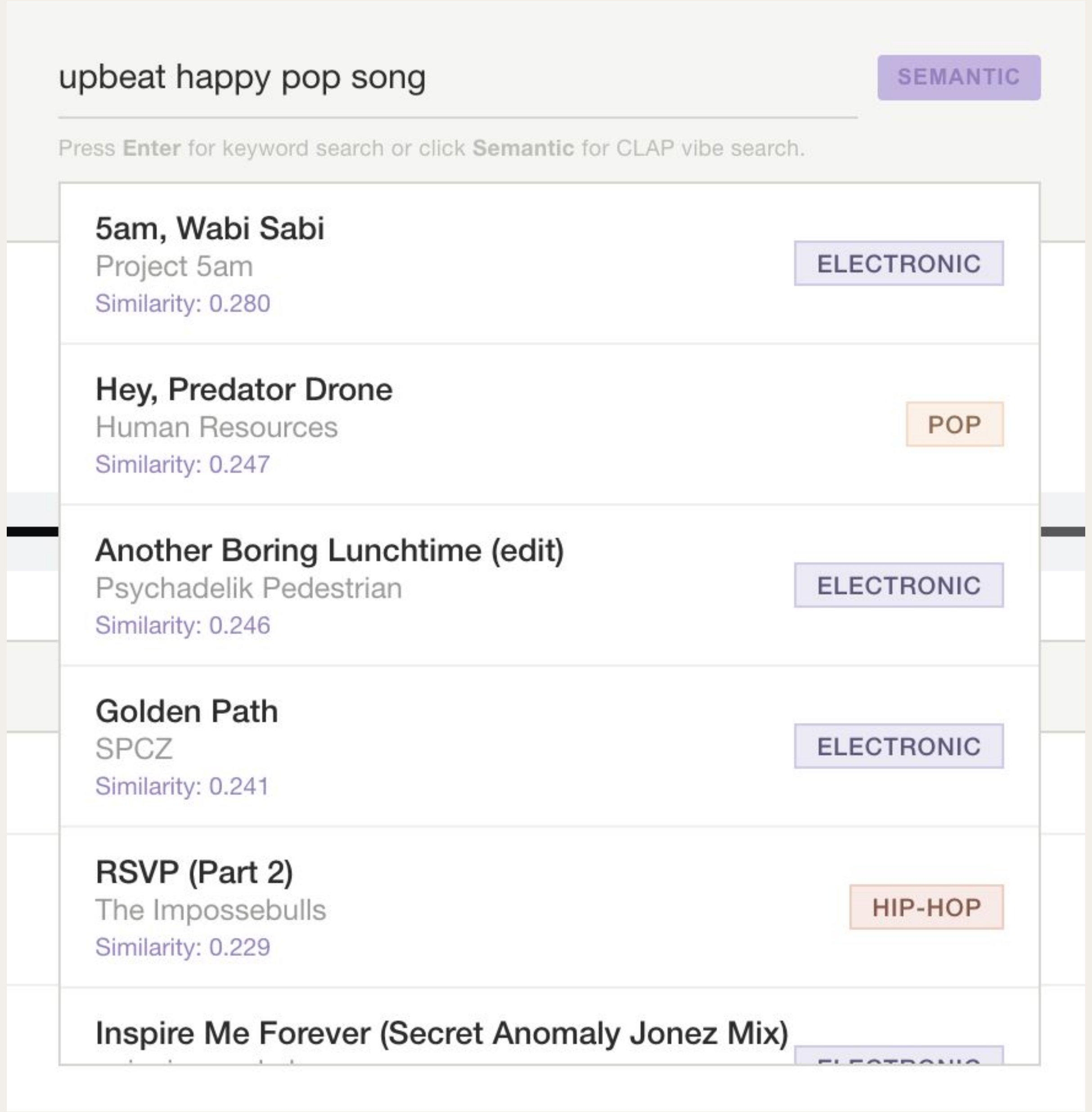
/ 01 — CLAP RESULTS

Natural-language queries against 2000 audio embeddings.

QUERY	TOP RESULT	GENRE	COS
"sad piano ballad"	Blue Dot Sessions	Instrumental	0.36
"aggressive heavy metal"	Lately Kind of Yeah — 恐怖 (Terror)	Electronic	0.42
"upbeat happy pop song"	Project 5am — 5am, Wabi Sabi	Electronic	0.28
"acoustic guitar folk"	Blue Dot Sessions — Wistful	Instrumental	0.40

The model correctly identifies "Instrumental" as the best fit for piano and acoustic guitar queries within this subset, **though "Electronic" is surfaced for the heavy metal query, indicating a limitation in the subset's coverage of specific high-intensity genres.**

CLAP model (Contrastive Language-Audio Pretraining) prioritizes acoustic texture over metadata labels.



/ SECTION 02

SBERT.

Lyrics & semantic search — a siamese transformer mapping metadata + lyrics to 384-d embeddings.

/ 02 — SBERT ANALYSIS

The core idea

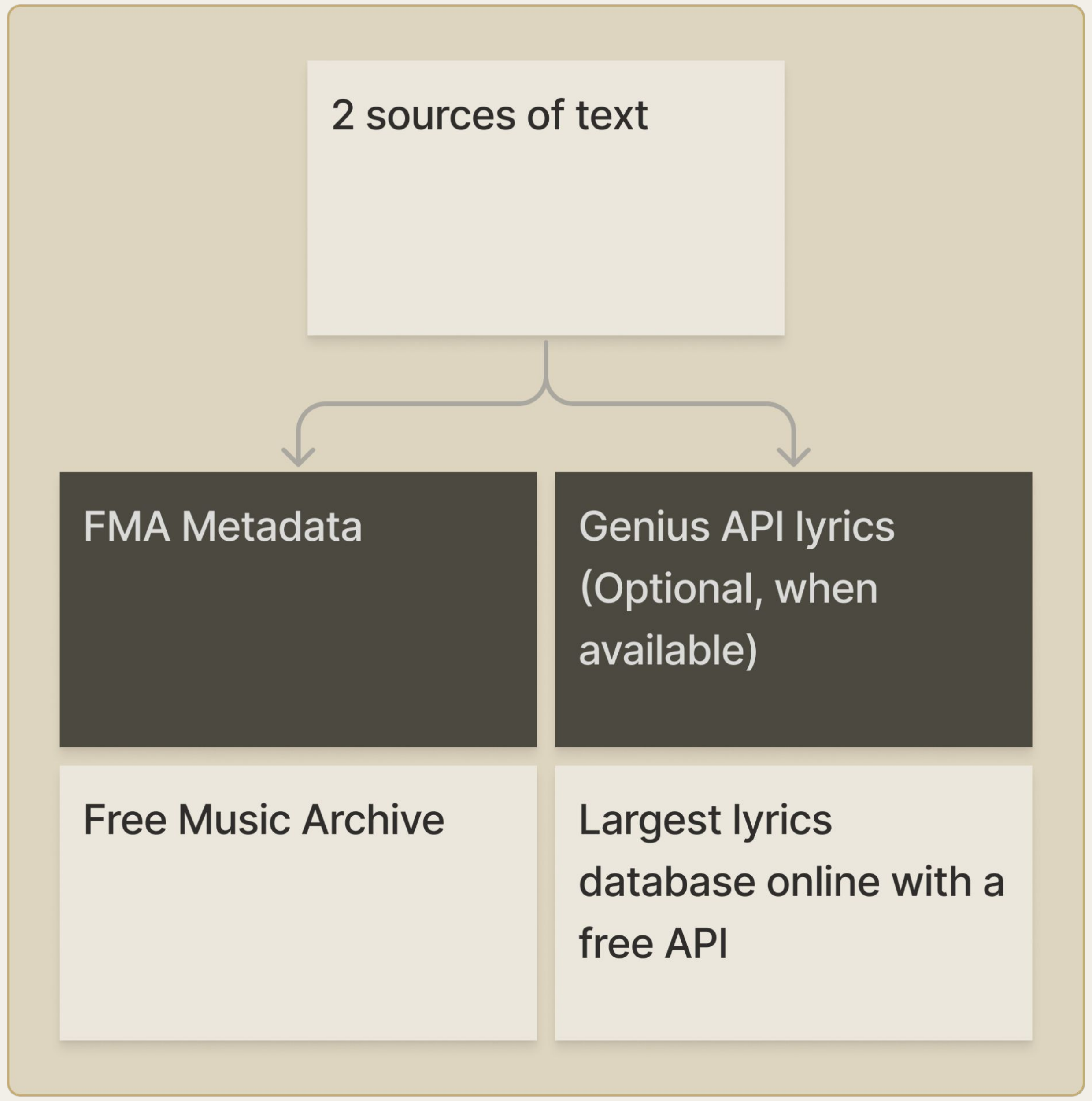
We are building a text-based view of each song. Instead of listening, we read everything written about the song and turning that into a searchable vector.

Sentence-BERT (2019)

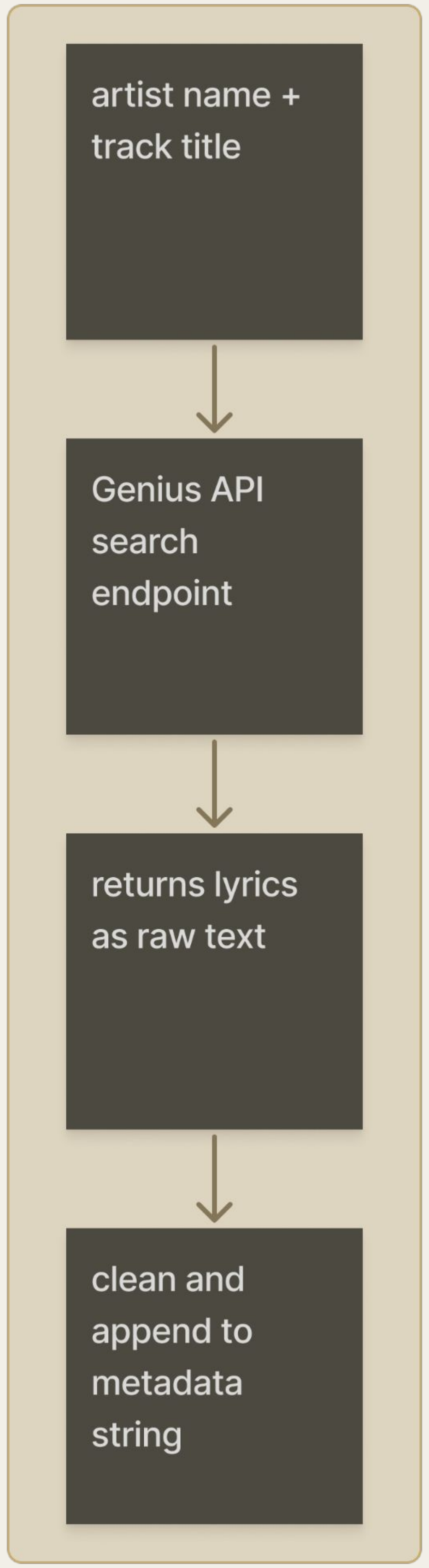
Instead of comparing sentences together, embed each sentence independently into a fixed vector

all-MiniLM-L6-V2 (2021)

Trained on a diverse range of multi-domain text data, compressed into a distilled version of a larger model



Genius API Flow



/ 02 — SBERT ANALYSIS

The complete pipeline

FMA tracks.csv → Build metadata string per track → Embed Genius API, append lyrics if found → **SBERT** (all-MiniLM-L6-V2) embeds each document → 384-dim vector per track → Store all 2000 vectors in FAISS index → User query → SBERT embed → FAISS search → top-K tracks → **Ranked List**

```
track_id: 2
title: “Death with Dignity”
artist: “Sufjan Stevens”
album: “Carrie & Lowell”
Genres: [“Folk”, “Indie folk”, “singer-
songwritier”]
tags: [“melancholy”, “acoustic”, “fingerpicking”]
```

```
import lyricsgenius
genius = lyricsgenius.Genius("YOUR_API_TOKEN")
song = genius.search_song("Death with Dignity",
"Sufjan Stevens")
lyrics = song.lyrics

#append
full_text = f"Metadata string: {metadata_string} +
Lyrics: {lyrics}"
```

```
def build_metadata_string(track):
    return (
        f"{track['title']} by {track['artist']}"
        f"Album: {track['album']}. "
        f"Genres: {', '.join(track['genres'])} "
        f"Tags: {', '.join(track['tags'])}"
    )
```

FAISS
Facebook AI Semantic Search

Built by Meta AI Research team

Given a vector, FAISS finds the most similar vector in a large collection, fast

/ SECTION 03

OPENL3.

Acoustic similarity — self-supervised audio embeddings capturing timbre, rhythm, and texture.

/ 03 — OPENL3

A purely acoustic space — no semantic supervision.

/ MODEL

OpenL3 (content_type='music' , 512-d). Trained via self-supervised audio-visual correspondence on AudioSet — captures low-level acoustic properties without any label supervision.

/ KEY DIFFERENCE FROM CLAP

CLAP maps audio into a space shared with text — semantic

L3 maps audio into a purely acoustic space — sounds-alike

→ Two songs that sound similar are close — even across genres

USE CASE

"Find songs that sound like this one."

Index all 8,000 tracks via FAISS for instant nearest-neighbour search.

2,000	512	<1ms
TRACKS INDEXED	DIMENSIONS	PER QUERY

/ 03 — OPENL3 / IMPLEMENTATION

How I built the embedding pipeline.

/ PIPELINE

librosa load (22,050 Hz) → openl3.get_audio_embedding() → mean pool frames → 512-d vector per track → FAISS IndexFlatIP

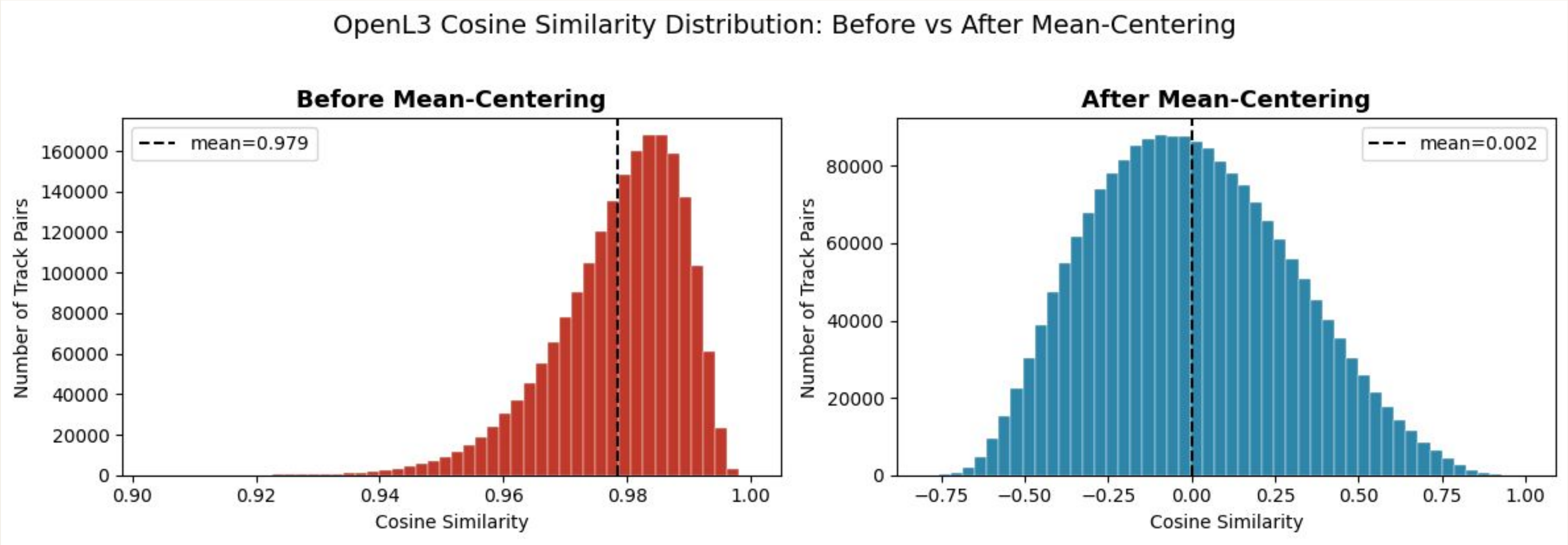
2,000 TRACKS PROCESSED

512 EMBEDDING DIMS

100 TRACK CHECKPOINT

/ DC OFFSET PROBLEM

RAW	Raw cosine similarities cluster artificially near 0.98 — a DC offset in the embedding space makes all tracks look similar
FIX	Mean-center embeddings before L2 normalization — removes the offset and makes similarity scores meaningful
→	Similarity scores now spread meaningfully from 0.3–0.9 across track pairs



CHECKPOINT SYSTEM

Colab sessions time out mid-run.

Progress saved to .npy every 100 tracks. On crash: resume from last checkpoint, skip done_set. No track processed twice.

/ SECTION 03

Evaluation

/ 04 — WHY FUSION/Cross-View Overlap

Fusion Method: RRF

/ INTERPRETATION

RRF calculates a **final score** for each document using the formula:

`score(d) = Σ 1 / (k + rank(result(q), d))`

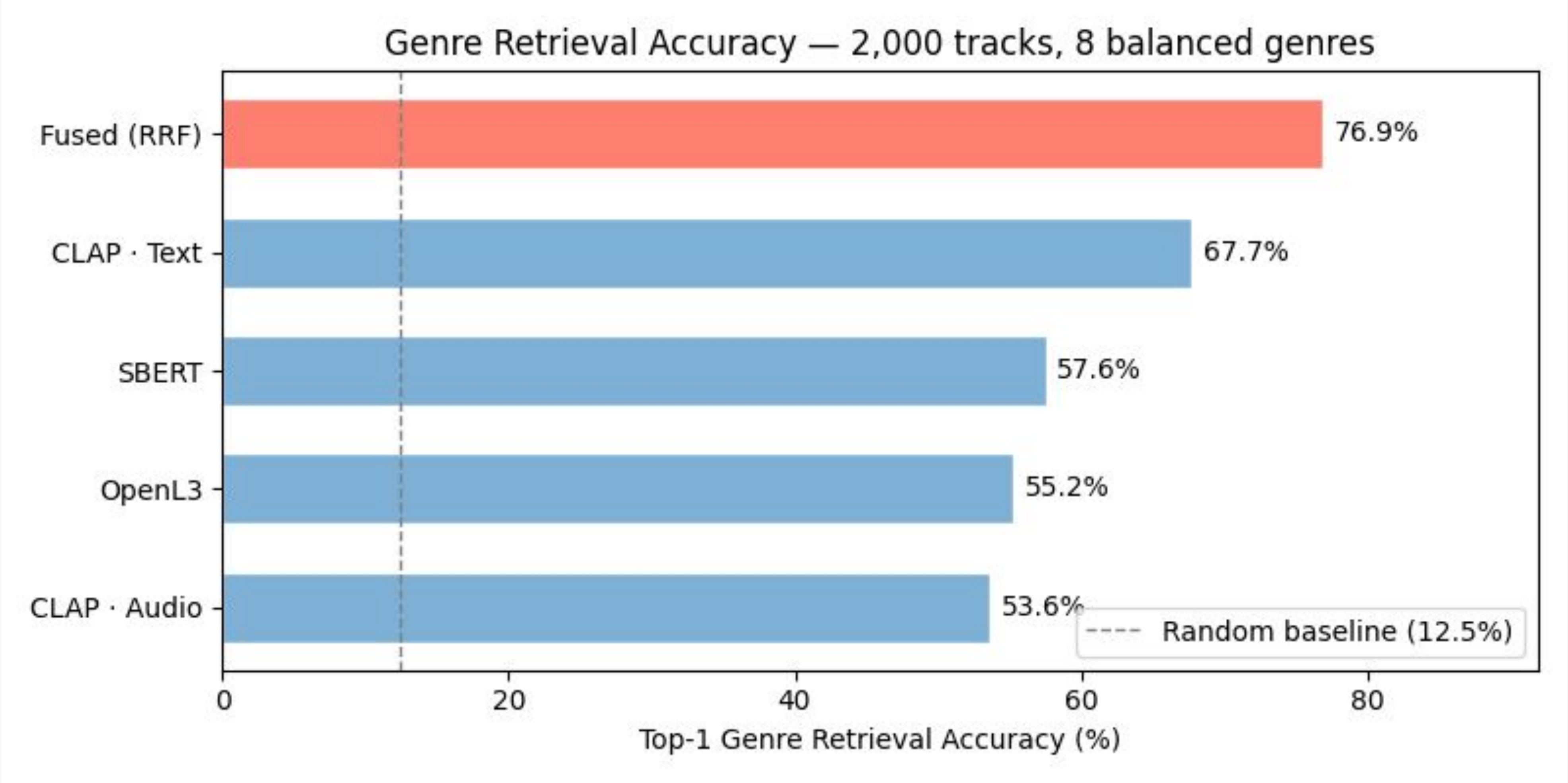
Where:

- `d` is a document in the result set
- `result(q)` is the ranked list of documents for query `q`
- `rank(result(q), d)` is the position of document `d` in that list
- k is a constant (=60) that smooths the influence of rank differences

Documents that appear high in multiple lists accumulate higher scores, while documents appearing in only one list receive lower scores. This approach **requires no training data** and is insensitive to the quality variations of individual input lists.

/ 04 — EVALUATION

Genre Retrieval Accuracy: Supervised ground truth — 2,000 tracks, 8 balanced genres, top 1 recommendation



/ KEY TAKEAWAYS

01 CLAP text embedding is strongest single view

02 Text semantics > acoustic features for genre

03 Fused model performs significantly better than four models alone

/ 04 — EVALUATION

Echo Nest Feature: Independent ground truth — 294 tracks the models never saw, top 5 recommendations

For each track, grab its top-5 nearest neighbors from each model, then measure how far apart they are in Echo Nest feature space. .
Lower distance = the recommendations are more acoustically similar in ways the model didn't directly learn.

METHOD	AVG DIST.	VS. RANDOM
Random baseline	3.80	—
SBERT	3.33	−12.4%
Fused · SBERT+OpenL3+CLAP	**2.87**	**−24.3%**
OpenL3 & CLAP	**2.87**	**−24.3%**

/ KEY TAKEAWAYS

01 OpenL3 & CLAP wins, because they have audio-related information.

02 Fused keeps the same with OpenL3 & CLAP, but is more robust (covering more tracks)

03 SBERT score is higher (larger distance) because it does not contain acoustic information to make sense of Echo Nest audio features (danceability, energy, valence, tempo, etc.)

/ SECTION 03

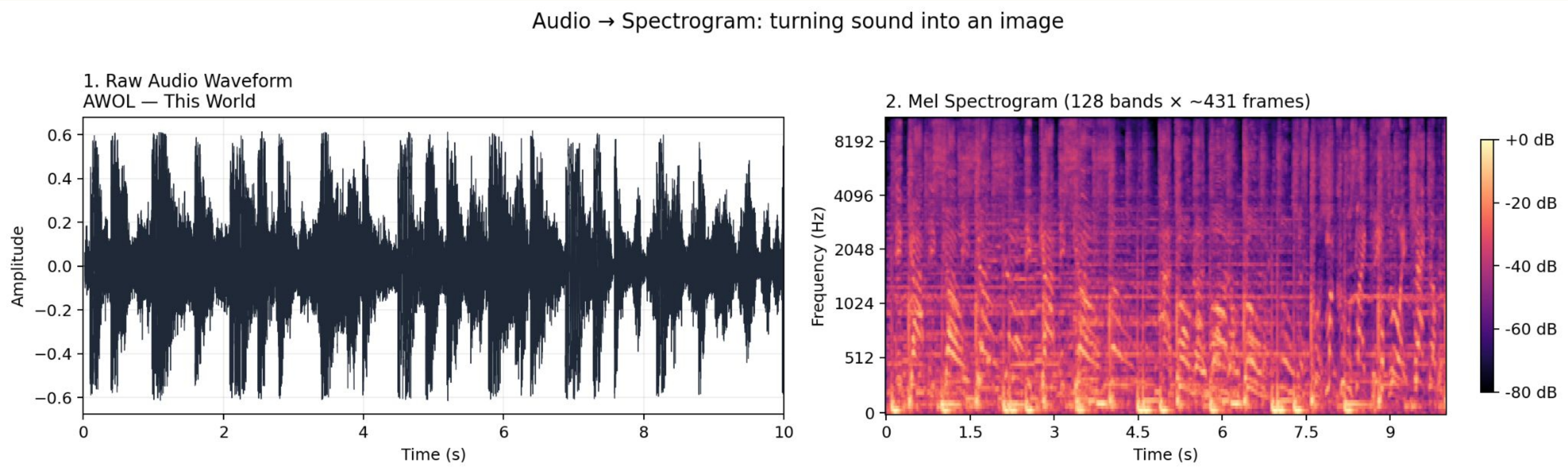
Other Experiment

Spectrogram analysis

/ 11 — Experimental

Experiments

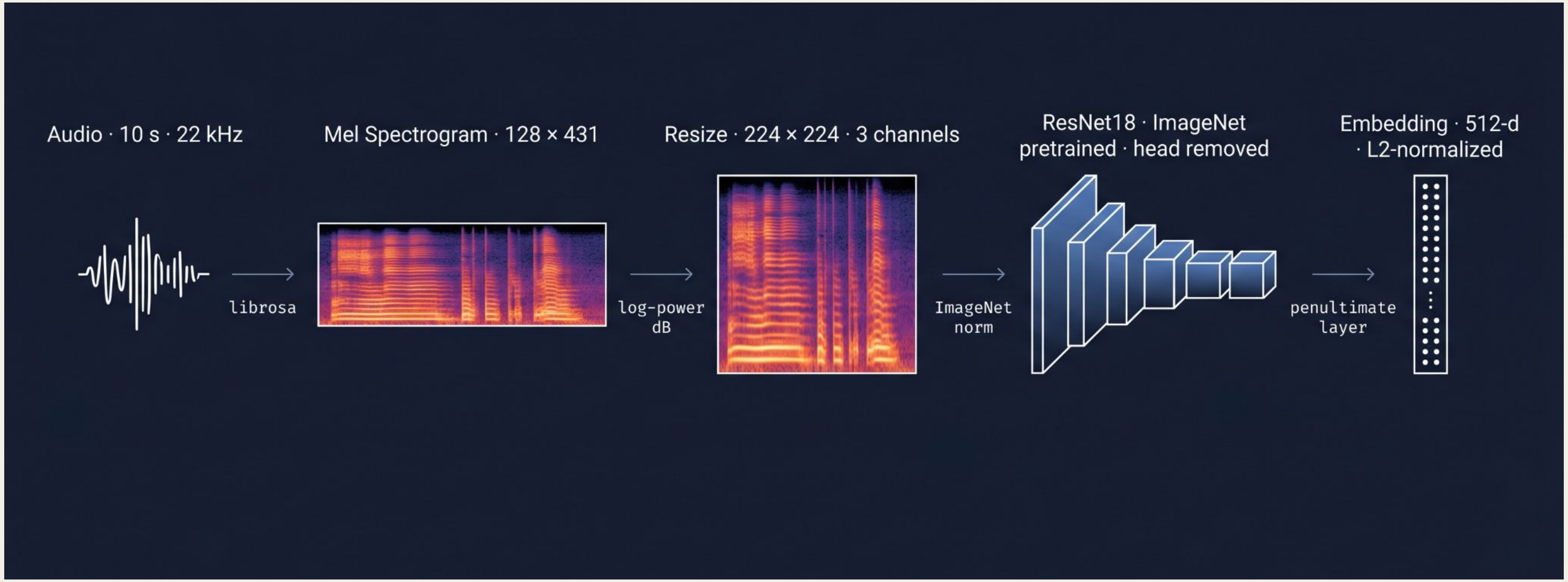
Can a CNN trained on photos still learn something useful from spectrograms, because they look like images?



/ 11 — Experimental

Feature-extraction pipeline

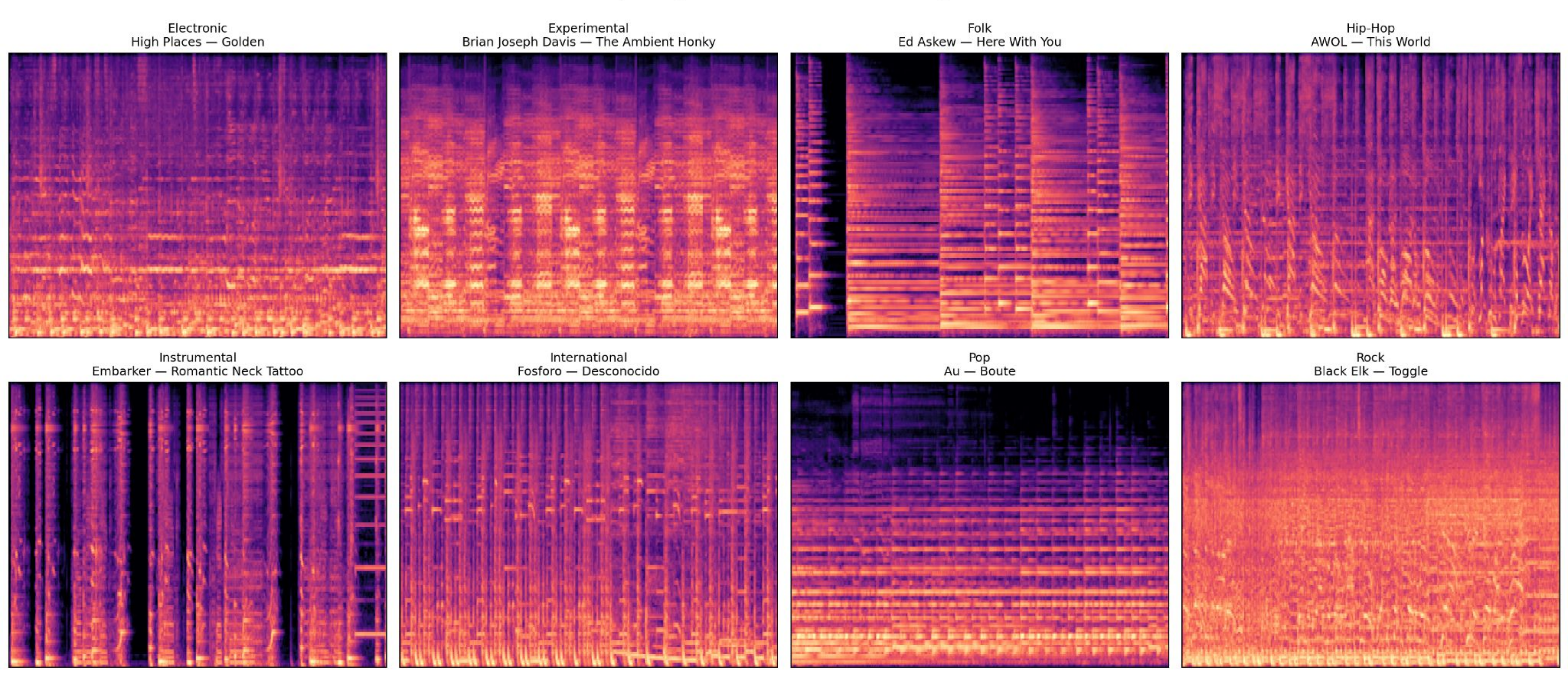
Visual features learned from natural images should still capture useful structure in spectrograms — harmonic stripes, drum hits, rhythmic patterns — without any audio-specific training.



Spectrogram

/ 11 — Experimental

Spectrogram per music genre



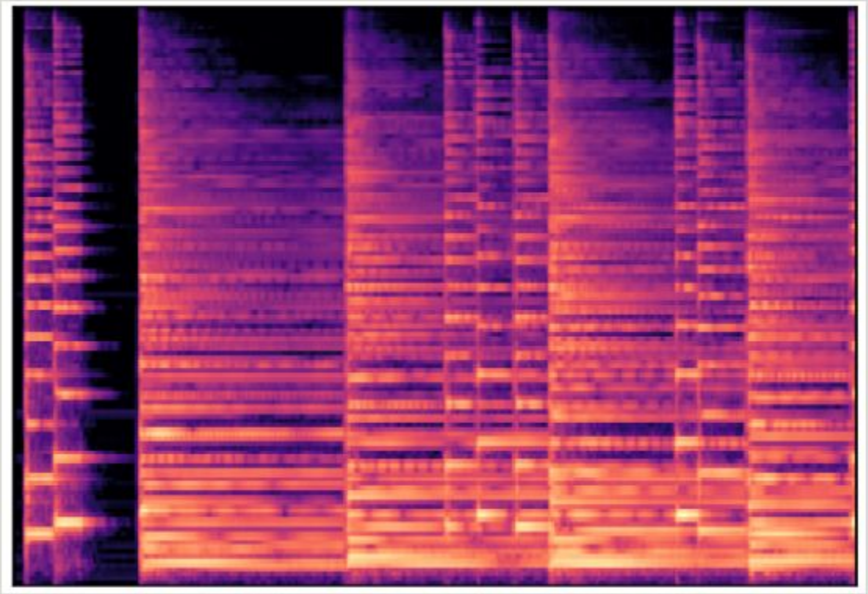
LIVE DEMO

WHERE SPECTROGRAM IS STRONG

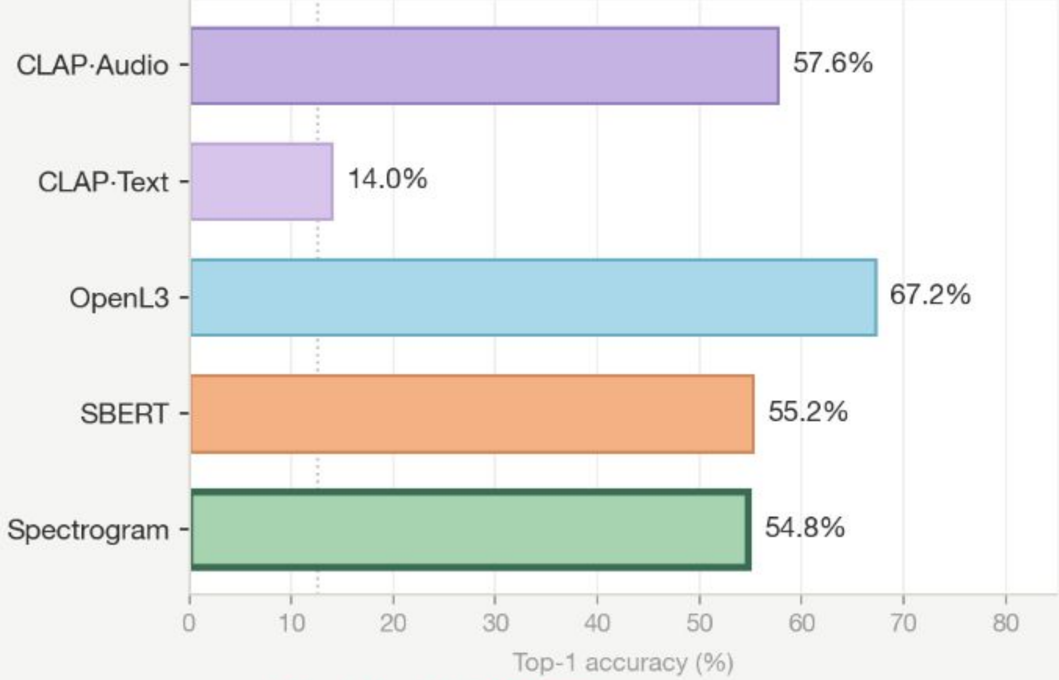
ImageNet visual priors transfer to genres with regular spectral structure.

Top-1 retrieval accuracy on three genres where Spectrogram matches or beats peer views —
all three share strong, repeating visual patterns the CNN's ImageNet filters can already encode.

FOLK

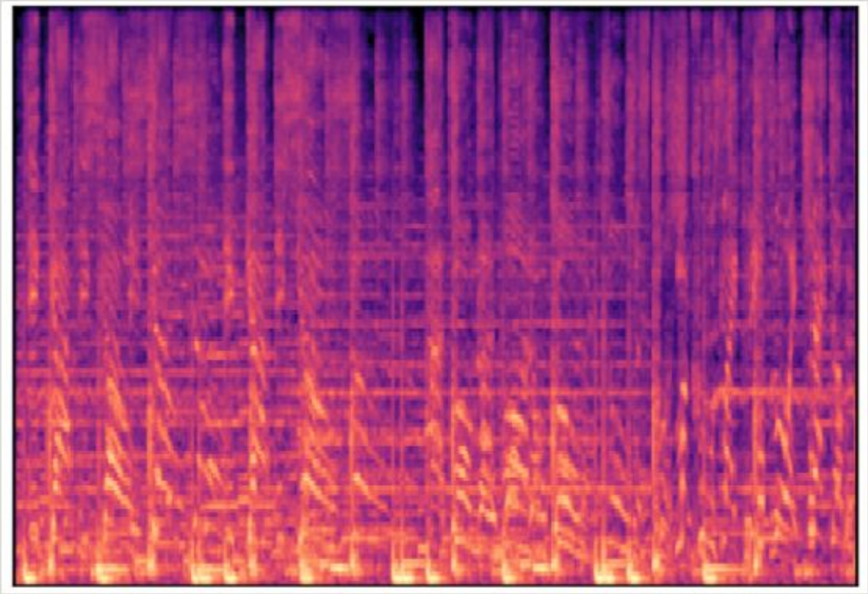


Horizontal harmonic stripes
from voice + acoustic guitar

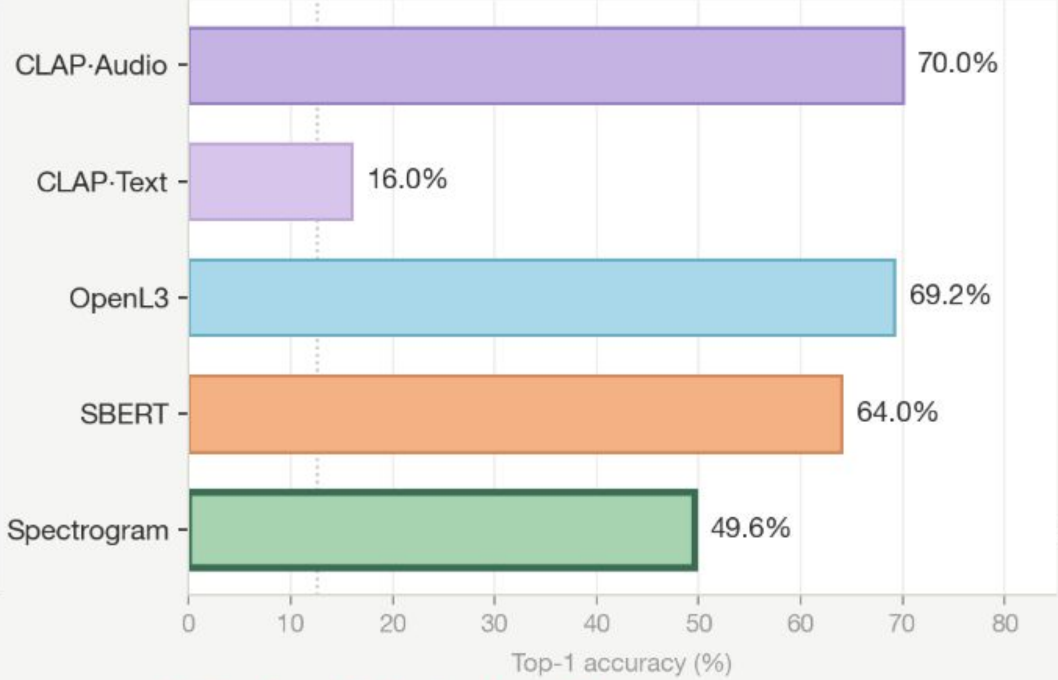


→ matches OpenL3 / SBERT

HIP-HOP

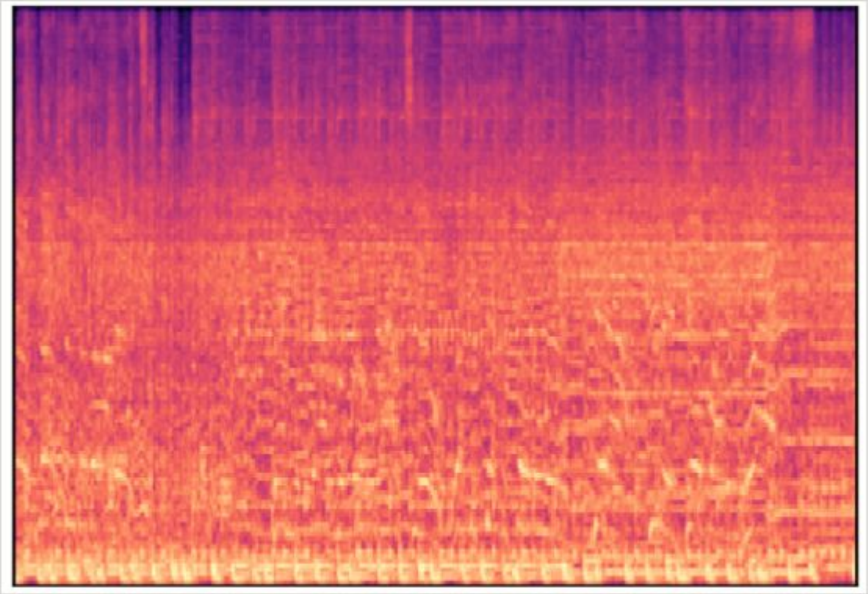


Regular vertical bands
from kick + snare grid

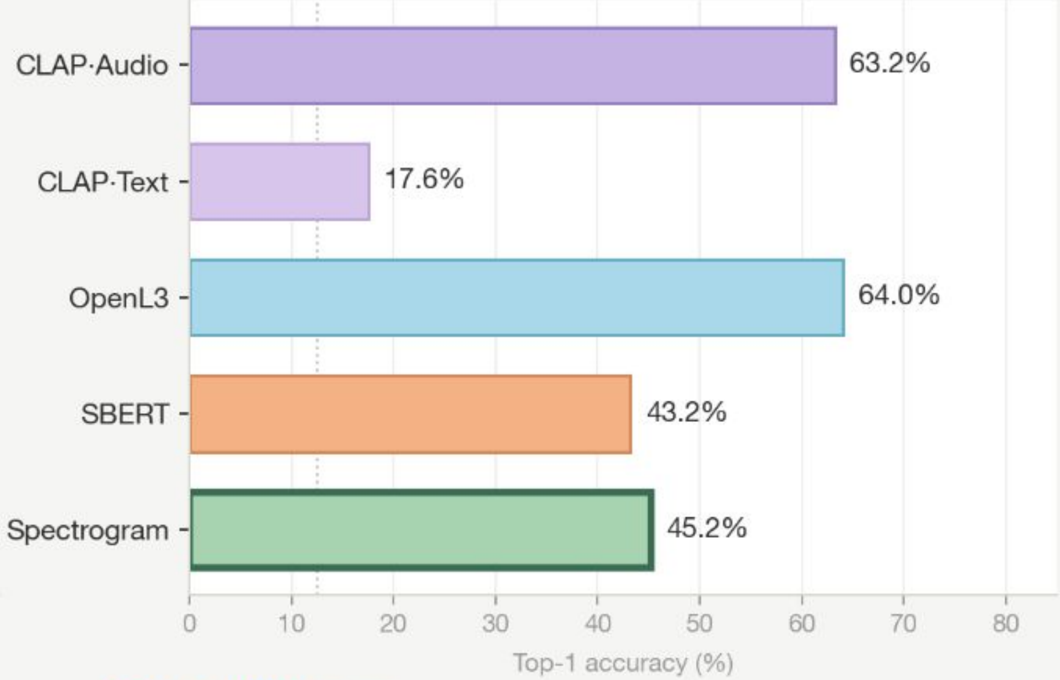


→ trails only audio-trained models

ROCK



Dense broadband texture
from distortion + drums



→ BEATS SBERT (text)

Spectrogram view · ResNet18 ImageNet-pretrained, frozen · 512-d · 2,000 FMA tracks · n=250 per genre · dotted line: random baseline 12.5%

/ THANK YOU

QUESTIONS?

/ DEMO

/ SOURCE

github · multi-faceted-music-retrieval

/ TEAM

Wenny · Sid · Issac · MJ · Jiayi · Helena

/ APPENDIX B — REFERENCES

Datasets, models & libraries.

/ DATASETS

FMA **Free Music Archive (Small)** 8,000 tracks · 8 genres · ~30s clips · open-licensed. Defferrard et al., 2017 · github.com/mdeff/fma

ECHO **Echo Nest / Spotify audio features** Danceability · energy · valence · tempo · acousticness · … 294-track FMA overlap → independent ground truth

GEN **Genius API** Song lyrics (first 1,000 chars per track).

AS **AudioSet** Training data for OpenL3 (audio-visual correspondence). Gemmeke et al., 2017 · research.google.com/audioset

/ MODELS & LIBRARIES

CLAP **LAION CLAP** · HTSAT-tiny backbone, AudioSet+music pretraining
laion/clap

SBERT **Sentence-Transformers** · all-MiniLM-L6-v2 · 384-d
sentence-transformers

L3 **OpenL3** · self-supervised audio embeddings
marl/openl3

FAI **FAISS** · Facebook AI Similarity Search
facebookresearch/faiss

/ APPENDIX A — GLOSSARY

Reference terms used throughout.

EMBEDDING
A fixed-size numerical vector representing a track in a continuous space where similar items are close.

FAISS
Facebook AI Similarity Search — efficient nearest-neighbour search. We use IndexFlatIP (exact inner product).

COSINE SIMILARITY
Cosine of the angle between two vectors, [-1, 1]. Equals dot product for L2-normalised vectors.

CONTRASTIVE LEARNING
Learns to map similar pairs close, dissimilar pairs far. Typically uses InfoNCE loss.

CLAP
Contrastive Language-Audio Pretraining — shared audio/text space; mood, genre, instruments.

SBERT
Sentence-BERT — siamese transformer for sentence embeddings. all-MiniLM-L6-v2.

OPENL3
Self-supervised audio embeddings on AudioSet — timbre, rhythm, texture (no labels).

RRF
Reciprocal Rank Fusion — sum 1/(k+rank) across views. Robust, no weight tuning.

DATA LEAKAGE
Eval targets leaking into model inputs, inflating metrics. We removed genre from SBERT input.

T-SNE
t-distributed Stochastic Neighbor Embedding — 2D visualisation of high-dimensional vectors.

INFONCE
Contrastive loss using softmax over (positive vs many negatives). Used for CLAP fine-tuning.

The Core Concepts

Q: What is an embedding? A: It’s a "digital fingerprint" that turns a track into a list of numbers so the computer can plot it on a map where similar-sounding songs sit close together.

Q: What is contrastive learning? A: A training method where the model learns by trying to pull "matching" pairs closer together while pushing "mismatching" pairs further apart.

Q: What is cosine similarity? A: A way to measure how similar two tracks are by looking at their "vibe" (angle) rather than their intensity (magnitude).

Q: What is data leakage? A: It’s "accidental cheating" where the answer to the test (like a genre label) gets mixed into the study guide, making the model look smarter than it actually is.